Exact value triangles

1

a $\frac{1}{2}$

b $\frac{\sqrt{3}}{2}$

c $\frac{1}{\sqrt{2}}$

d $\frac{1}{\sqrt{2}}$

e $\frac{1}{2}$

f $\frac{\sqrt{3}}{2}$

g $\frac{\sqrt{3}}{1}$

h 1

i $\frac{1}{\sqrt{3}}$

2

a 1

b $\frac{\sqrt{6}}{4}$

c $\frac{\sqrt{3}+\sqrt{2}}{2}$

d $\frac{\sqrt{2}}{4}+1$

Unknown sides

3

a $h=1$

b $x=2$

c $w=28$

d $x=\frac{\sqrt{30}}{2}$

4

a $a=\frac{7\sqrt{3}}{2}$

b $b=\frac{7}{2}$

5

a

i $x=3\sqrt{6}$

ii $y=3\sqrt{3}$

b

i $x=2\sqrt{6}$

ii $y=2\sqrt{2}$

6

a $m=5$

b $t=6+5\sqrt{3}$

Unknown angles

7

a $\theta=60^{\circ}$

b $\theta=60^{\circ}$

c $\theta=45^{\circ}$

d $\theta=30^{\circ}$

8 $\theta=45^{\circ}$

9

a $\theta=30^{\circ}$

b $\frac{1}{\sqrt{3}}$

10

a $\theta = 45^\circ$

b $\theta = 60^\circ$